

Invited paper

Synthesis of non-peripheral thioanisole-substituted phthalocyanines:
Photophysical, electrochemical, photovoltaic, and sensing properties

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ABSTRACT

This work reports on the synthesis, characterization, photophysical, electrochemical, photovoltaic and sensing properties of non-peripherally tetra 4-(methylthio)phenoxy substituted metal-free (**2**), Zn(II) (**3**) and Co(II) (**4**) phthalocyanine derivatives. Confirmation of the synthesized phthalocyanine structures which have the good solubility in organic solvents, performed with a combination of elemental analysis, FTIR, ^1H NMR, ^{13}C NMR, UV–vis and MALDI-MS spectral data. Electrochemical measurements exhibit that the complex (**4**) shows the redox reactions taking place on both metal center and the ring while the phthalocyanines (**2**) and (**3**) give only ring-based redox reactions. All the complexes were examined for utilization in dye-sensitized solar cells (DSSCs) in the presence of chenodeoxycholic acid and the highest power conversion efficiency was achieved for the complex (**3**), as 1.09%. The interaction of phthalocyanines (**2**) and (**3**) with Ag^+ and Pd^{2+} ions and their aggregation properties (H- or J-Aggregation) were investigated by UV–vis spectroscopy. Photophysical (fluorescence quantum yield) properties of the complexes (**2**) and (**3**) were reported in different solvents, such as tetrahydrofuran (THF) and dimethyl sulfoxide (DMSO). The results of spectral measurements in these solvents showed that Φ_F value of the complexes (**2**) and (**3**) were found to be lower than the unsubstituted ZnPc.

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1. Introduction

Metallophthalocyanine complexes (MPcs) have attracted great attention in recent years because of their excellent redox properties used in different fields such as non-linear optics, solar cells, molecular electronics, optical storage devices, electrocatalysts, sensing elements in gas and chemical sensors, electrochromic display, organic field effect transistors and photosensitizers in photodynamic cancer therapy [1]. The main target for researchers is to synthesize Pcs having good solubility by attaching various substituents on Pcs ring, such as alkyl, alkoxy, phenoxy, macrocyclic groups and/or inserting the different central metal ions into the inner phthalocyanine core [2,3] because of very low solubility of unsubstituted phthalocyanines in water and organic solvents. On the other hand, tetra-substituted Pcs exhibit more solubility than that of octa-substituted Pcs because of the unsymmetrical arrangement of substituents in the latter one [4]. Moreover, attaching some functional groups at non-peripheral region of Pcs causes to red shifting of the Q-band and diminishes aggregation as regards and the binding of electron donating

sulphur groups at the peripheral region of Pc gives rise to the red shifting of the Q-band to more longer wavelengths, suggesting the varied properties that is important to be used in many fields [5,6].

The performances of MPcs are based mainly on their redox processes which occur on the Pc ring and the metal center. For example, the redox processes occur only at the ring in the Pcs with redox-inactive metal center, such as Zn and Cu, whereas in the presence of redox-active metal centers like Co and Mn, the electrochemical processes occur both at the metal center and at the ring [7–9]. The understanding of the electrochemical properties of the Pc complexes is important for their potential applications in many fields [10]. For example, utilization of Pcs in solar cells is highly fascinating because of their excellent photophysical and electrochemical properties [11]. At present, maximum power conversion efficiencies over 6% have been achieved for dye-sensitized solar cells (DSSCs) with both symmetric [12] and asymmetric Pcs [13]. However, the Pcs for solar cells were mainly asymmetric structures were attached with complicated anchoring groups, such as carboxylic acid or pyridine. Although the conversion efficiencies of symmetric Pcs are generally low, symmetric Pcs are low-cost, facile synthesis, high yield, and good solubility compared with asymmetric Pcs [14,15].

As also known, substituted Pcs can create two kinds aggregates, such as H-aggregates (face-to-face) and J-aggregates (head-to-

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tail). They influence their electronic and optical properties. The entity of H-type aggregation can be proved by the decreasing of intensity of the Q-band (monomeric species) and then the appearance of a new broader blue-shifted band (aggregated species). On the other hand, if there is a new broader red-shifted band on the higher wavelengths, this indicates J-type aggregation [16].

In this study, symmetrical and soluble 4-(methylthio)phenol non peripherally substituted metal-free (**2**), Zn(II) (**3**) and Co(II) (**4**) Pcs were designed/synthesized and characterized. The electrochemical results revealed that the redox reactions of all the Pcs are reversible or quasi-reversible processes. The electrochemical, optical and photovoltaic results demonstrated that the only complex **3** can be considered as an alternative sensitizer for DSSCs.

The sensing behavior and aggregation of these Pcs (**2**) and (**3**) in the presence of Ag^+ and Pd^{2+} soft metal ions have been studied by UV–vis spectroscopy, from these results, we have proved that, the interaction of the complex (**2**) with both Ag^+ and Pd^{2+} soft metal ions caused the formation of H-type aggregation, but differently there wasn't observed any aggregation/sensitivity towards Ag^+ metal ions for the complex (**3**) while showing the sensitivity to Pd^{2+} soft metal ions suggesting the formation of J-type aggregation. In addition, photophysical properties have been discussed such as, the effect of different solvents (THF and DMSO) on fluorescence quantum yields of the complexes (**2**) and (**3**).

2. Experimental

2.1. Materials and methods

Tetrahydrofuran (THF), chloroform (CHCl_3), 4-(methylthio)phenol, 3-nitrophthalonitrile, $\text{Zn}(\text{CH}_3\text{COO})_2$, CoCl_2 and all other reagents were purchased from Alfa Aesar, Merck, Aldrich, Fluka and used with no purification. The purity of the products was controlled in all step by TLC (SiO_2 , $\text{CHCl}_3/\text{MeOH}$ and THF/MeOH). UV–vis spectra was ace:inf>), 4-(methylthio)phenol, 3-nitrophthalonitrile, $\text{Zn}(\text{CH}_3\text{COO})_2$, CoCl_2 and all other reagents were purchased from Alfa Aesar, Merck, Aldrich, Fluka and used with no purification. The purity of the products was controlled in all step by TLC (SiO_2 , $\text{CHCl}_3/\text{MeOH}$ and THF/MeOH). UV–vis spectra was acquired in a quartz cuvette on an Agilent Model 8453 diode array spectrophotometer. Perkin-Elmer spectrum two FT-IR spectrophotometer was used to take FT-IR spectra. All the products were purified by chromatography on silica gel (Merck grade 60) from Aldrich. All reactions were achieved under a dry N_2 atmosphere. A Bruker 300 spectrometer instruments was used to take ^1H NMR and ^{13}C NMR spectra. Fluorescence measurement were recorded by Hitachi S-7000 fluorescence spectrophotometer. Mass spectras (MS) were analyzed by MALDI SYNAPT G2-Si Mass Spectrometry.

2.2. Electrochemical measurements

Cyclic voltammetry (CV) and square wave voltammetry (SWV) analyses were performed with a PARSTAT 2273 electrochemical analyzer (Princeton Applied Research) using a three-electrode cell with a glassy carbon working electrode (GCE, 3 mm diameter), a Pt wire counter electrode, and a non-aqueous Ag/AgCl reference electrode. The electrolyte consisted of 0.1 M tetrabutylammonium tetrafluoroborate (TBAF_4 , Merck) solution in dichloromethane (DCM) containing 0.5 mM of the dye. The solution was purged with high purity argon prior to measurement and blanketed with argon during measurement. In order to calibration of the reference electrode, electrochemical properties of the ferrocene/ferrocenium (Fc/Fc^+) redox couple as an external reference was performed under our experimental conditions. The half-wave peak potential ($E_{1/2}$), the peak separation potential (ΔE_p), ratio of anodic to

cathodic peak current (I_{pa}/I_{pc}) of Fc/Fc^+ were 0.54 V, 109 mV, and ~ 1.00 at scan rate of 0.100 V s^{-1} , respectively. The ΔE_p value is higher than 59 mV which is expected for reversible electrochemical process of Fc/Fc^+ . However, non-ideal electrochemical behavior of Fc/Fc^+ was reported previously [17].

2.3. Solar cell fabrication and characterization

Fluorine-doped tin oxide (FTO) glass plates (Solaronix, TC022–15, $15 \Omega \text{ cm}^{-2}$, >80% transmittance in the wavelengths between 400 and 700 nm) were washed with water and ethanol. Then, the plates were dipped in 40 mM TiCl_4 aqueous solution at 50°C for 30 min and washed with water. A transparent layer of TiO_2 nanoparticles (Solaronix, Ti–Nanoxide T/SP) was prepared by the doctor blade method on FTO plates, which was then gradually heated to 500°C and baked at this temperature for 30 min under ambient condition. A home-made scattering layer of TiO_2 nanoparticles was synthesized according to the published procedure [18] and deposited by using the same method for the former layer on the treated FTO plates after that these plates cooled up to room temperature. Then, it was gradually heated to 500°C for 30 min again. Finally, the double layer of TiO_2 coated plates were again treated with the TiCl_4 at 50°C for 30 min and sintered at 500°C for 30 min. The active areas of the photanodes were controlled at $\sim 0.16 \text{ cm}^2$ with adhesive tape. After cooling down to room temperature, the dye-sensitized photoanodes were obtained by dipping the coated plates into the each Pc solution (0.3 mM) of ethanol/tetrahydrofuran (2/1, v/v) solution containing various chenodeoxycholic acid (CDCA) concentrations at ambient temperature for 72 h. The Pt counter electrodes obtained by a thermal deposition of Pt paste (Solaronix, Platisol T) on the surface of FTO plates at 500°C for 30 min. The dye-sensitized photoanodes and the Pt counter electrodes were assembled in sandwiched type cell using a spacer (Solaronix, Meltonix 1170–25). The electrolyte solution (Solaronix, Iodolyte AN–50) was injected into the interface between the photoanode and counter electrode.

The DSSCs were irradiated under light irradiance of 100 mW cm^{-2} (AM1.5) from a 150 W solar simulator (Newport, 96000). The above-mentioned potentiostat/galvanostat was used to obtain the current-voltage (J - V) curves. The intensity of the incident light was calibrated using a standard Si detector (Newport, 918D-SL-OD3) and an optical power meter (Newport, 1917-R). In order to confirm the reproducibility of the cells, three J - V measurements were performed for each synthesized Pc and it was found that the dispersion is lower than 8% for each DSSC.

2.4. Synthesis

2.4.1. 3-(4-(methylthio)phenoxy)phthalonitrile (**1**)

4-(methylthio)phenol (0.809 g, 5.78 mmol) was dissolved in 10 cm^3 dry DMF and then finely dried K_2CO_3 ($\sim 2.00 \text{ g}$, excess) was added to this mixture. Then 3-nitrophthalonitrile (1.00 g, 5.78 mmol) was added to this mixture dropwise by stirring effectively at 40°C under the N_2 atmosphere. The reaction mixture was kept for 3 day at this temperature under N_2 atmosphere. After the reaction mixture was cooled to room temperature, it was poured into ca. 200 cm^3 ice-water media. The occurred precipitate was filtered and then washed with ca. 100 cm^3 water up to the washings became neutral. The occurred precipitate was filtered and dissolved in CHCl_3 and washed with 5% NaHCO_3 to eliminate the beginning residual compounds. The creamy solution was then treated with anhydrous Na_2SO_4 until become dry and filtered. It was purified by chromatography over a silica gel column by using a eluent that is a mix of CHCl_3 : MeOH (100/5), giving white powder, (**1**). Finally, the pure powder was dried in a vacuo.

Yield of **(1)**: 1.28 g (83%); m.p. = 119 °C; Anal. Calcd for $C_{15}H_{10}N_2OS$ (266 g mol⁻¹): C, 67.58; H, 3.52; N, 10.30. FT-IR (cm⁻¹): 3068 (w, Ar-CH), 2922 (w, Alip-CH), 2228 (C≡N, st), 1610 (C=C), 1567 (C=N), 1484 (st), 1273 (Ar-S-Alip-CH), 1088, 1009, 985, 847, 795. ¹H NMR (DMSO-*d*₆) δ: 8.11 (t, 1H, meta to Ar-O-Ar and CN), 7.88 (d, 1H, ortho to Ar-O-Ar), 7.52 (d, 1H, ortho to CN), 7.36–7.01 (m, 4H, O-Ar-S) 2.23 (s, 3H, CH₃-S-Ar). ¹³C NMR (DMSO-*d*₆) δ: 165.4, 156.7, 141.4, 141.1, 133.5, 133.4, 127.1, 126.3, 121.2, 121.1, 118.8, 110.2, 20.4. EI/MS *m/z*: 266.12 [M]⁺.

2.4.2. 1(4), 8(11), 15(18), 22(25)-Tetrakis(3-(4-(methylthio)phenoxy) metal-free phthalocyanine (2)

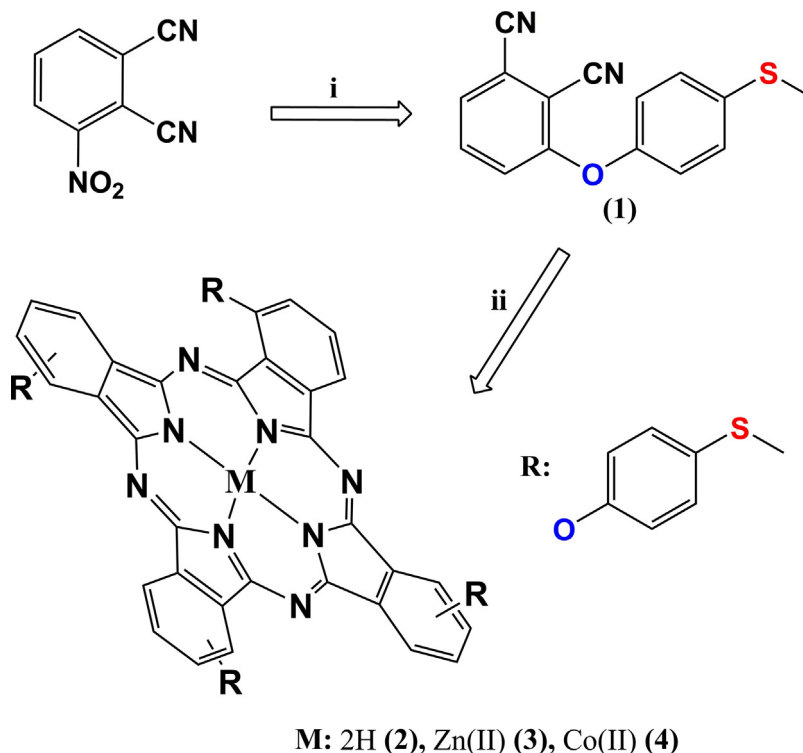
Compound **(1)** (0.150 g, 0.564 mmol) and 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU) (0.05 cm³) in hexanol (2.00 mL) was added in glass tube and heated with efficient stirring at 140 °C under N₂ atmosphere for 8 h. The color of the mixture changed into green-blue pending the course of time. The green-blue product was cooled to room temperature. After that, the mixture was washed with hexane. The crude product was washed several times successively with first MeOH, i-PrOH, and then with CH₃CN to eliminate residual impurities. Finally, the product was further purified by chromatography over a silica gel column by using a eluent that is a mix of [CHCl₃-THF (10/3 v/v)]. This product is soluble in CHCl₃, THF, DMF, DMSO, pyridine. Yield 0.020 g (14%), mp >200 °C. Anal. calcd. for C₆₀H₄₂N₈S₄O₄ (1067 g/mol): C, 67.52; H, 3.97; N, 10.50. Found: C, 67.45; H, 3.65; N, 9.97. FT-IR (cm⁻¹): 3286 (w, N—H), 3069 (w, Ar-CH), 2918–2849 (Alip-CH), 1615 (C=C), 1578 (C=N), 1484 (st), 1137, 1090, 955, 867, 749. ¹H NMR (DMSO-*d*₆): δ, ppm 8.82 (t, 4H, meta to Ar-O-Ar), 8.31 (d, 4H, ortho to Ar-O-Ar), 7.63 (d, 4H, para to Ar-O-Ar), 7.52–7.34 (m, 8H ortho to ArSCH₃), 7.24–7.16 (m, 8H meta to ArSCH₃) 2.24 (s, 12H, ArSCH₃). UV–vis (THF): λ_{max}/nm (Log ε) 686 (4.61) (Q_x), 715 (4.63) (Q_y), 626 (4.08) (n-π*, sh), 333 (4.38) (B). MS (MALDI-MS, 2,5-dihydroxybenzoic acid as matrix): 1068.78 [M+H]⁺.

2.4.3. 1(4),8(11),15(18), 22(25)-Tetrakis(3-(4-(methylthio)phenoxy) zinc (II) phthalocyanine(3)

Compound of **(1)** (0.150 g, 0.564 mmol) and 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU) (0.05 cm³) in hexanol (2.00 mL) was added in glass tube and heated with efficient stirring at 140 °C for 0.5 h under inert N₂ atmosphere, then finely dried Zn(CH₃COO)₂ (0.04 g, 0.22 mmol) was added. After heating and stirring for 8 h, the deep green-blue product was cooled to rt, and the solid was washed with hot hexane, propanol and cold acetonitrile to eliminate the residual impurities until the filtrate was clear. The green-blue product was purified by chromatography over a silica gel column by using a eluent that is a mix of [CHCl₃-THF (10/3 v/v)] and then dried in vacuo. The complex **(3)** is soluble in CHCl₃, CH₂Cl₂, DMF, DMSO, DMAA, and pyridine. Yield 0.038 g (24%), mp >200 °C. Anal. calcd. for C₆₀H₄₀N₈S₄O₄Zn (1131 g/mol): C, 63.68; H, 3.65; N, 9.90. Found: C, 63.25; H, 3.52; N, 9.81. FT-IR (cm⁻¹): 3057 (w, Ar-CH), 2920–2851 (Alip-CH), 1646 (C=C), 1577 (C=N), 1473 (st), 1244 (Ar-S-Alip-CH), 1087, 1045, 967, 879, 639. ¹H NMR (DMSO-*d*₆): δ, ppm 8.88 (t, 4H, meta to Ar-O-Ar), 8.32 (d, 4H, ortho to Ar-O-Ar), 7.56 (d, 4H, para to Ar-O-Ar), 7.54–7.37 (m, 8H ortho to ArSCH₃), 7.1–6.62 (m, 8H meta to ArSCH₃) 2.14 (s, 12H, ArSCH₃). UV–vis (THF): λ_{max}/nm (Log ε) 691 (4.65) (Q), 625 (4.04) (n-π*, sh), 337 (4.30) (B). MS (MALDI-MS, 2,5-dihydroxybenzoic acid as matrix): 1132.58 [M+H]⁺.

2.4.4. 1(4), 8(11), 15(18), 22(25)-Tetrakis(3-(4-(methylthio)phenoxy) cobalt (II) phthalocyanine(4)

Compound of **(1)** (0.150 g, 0.564 mmol) and 1,8-diazabicyclo [5.4.0]undec-7-ene (DBU) (0.05 cm³) in hexanol (2.00 mL) was added in glass tube and heated with efficient stirring at 140 °C for 0.5 h under N₂ atmosphere, then finely dried CoCl₂ (0.03 g, 0.13 mmol) was added. After heating and stirring for 8 h. After cooling to room temperature and hexane added. The dark green-blue crude product was then successively washed to eliminate



Scheme 1. Synthetic route of 1(4), 8(11), 15(18), 22(25)-Tetrakis(3-(4-(methylthio)phenoxy) phthalocyanine M: 2H(2), Zn(II)(3), Co(II)(4). i: K₂CO₃, 4-(methylthio)phenol, DMF, 40 °C, 2 days. ii: without metal salt, anhydrous Zn(CH₃COO)₂, CoCl₂ 140 °C, for 8 h DBU.

impurities with CH_3CN , isopropanol and dried. Further purification by chromatography over a silica gel column by using a eluent that is a mix of $[\text{CHCl}_3\text{-THF (10/3 v/v)}]$ and dried in vacuo. Blue phthalocyanine (**4**) is soluble in CHCl_3 , THF, MeOH, DMF, DMSO, DMSO, DMAA, and pyridine. Yield 0.030 g (20%), mp > 200 °C. Anal. calcd. for $\text{C}_{60}\text{H}_{40}\text{CoN}_8\text{S}_4\text{O}_4$ (1124 g/mol): C, 64.05; H, 3.67; N, 9.96. Found: C, 63.98; H, 3.20; N, 9.56. FT-IR (cm^{-1}): 3060 (w, Ar-CH), 2918–2850 (Alip-CH), 1615(C=C), 1579 (C=N), 1480(st), 1240 (Ar-S-Alip-CH), 1089, 975, 798, 758. UV-vis (THF): $\lambda_{\text{max}}/\text{nm}(\text{Log } \epsilon)$ 679 (4.75) (Q), 617(4.26) (n- π^* , sh), 322(4.59) (B). MS (MALDI-MS, 2,5-dihydroxybenzoic acid as matrix): 1125.38 $[\text{M}+\text{H}]^+$.

3. Result and discussion

3.1. Synthesis and spectroscopic characterizations

In this study, 3'-(4-(methylthio)phenoxy)phthalonitrile (**1**), its metal free(**2**), Zn(II)(**3**) and Co(II)(**4**) phthalocyanines were synthesized. Phthalocyanines (**2–4**), were achieved by heating a composition of 3-(4-(methylthio)phenoxy)phthalonitrile with dried $\text{Zn}(\text{CH}_3\text{COO})_2$, CoCl_2 and/or with no metal salt under N_2 atmosphere at ca. 140 °C in the existence of 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) for 8 h (Scheme 1). All the synthesized phthalocyanines were purified by column chromatography over a silica gel using a mixture of $[\text{CHCl}_3\text{-THF (10/3 v/v)}]$. The solubility of these new complexes in organic solvents such as CHCl_3 , THF, MeOH, DMF, DMSO, DMAA, and pyridine.

The results for the structure of phthalocyanines (**2**), (**3**) and (**4**) that obtained from Fourier transform infrared spectroscopy (FT-

IR), Proton and carbon-13 nuclear magnetic resonance (^1H and ^{13}C NMR), MALDI-MS and UV-vis spectroscopic methods, and also by elemental analysis.

The disappearance of the sharp $\text{C}\equiv\text{N}$ vibration at 2228 cm^{-1} confirmed the existence of cyclotetramerization of 3'-(4-(methylthio)phenoxy)phthalonitrile to the phthalocyanines (**2–4**). The FT-IR spectra of phthalocyanines (**2–4**) are almost identical except for small stretching shifts. The characteristic vibrations corresponding to C=C groups at around $\sim 1615\text{--}1646\text{ cm}^{-1}$ and aromatic CH stretching around $3057\text{--}3069\text{ cm}^{-1}$ were observed for all compounds. Aliphatic CH stretches were observed around $2849\text{--}2920\text{ cm}^{-1}$ for the methyl groups. The appearance of the inner cavity NH stretches at 1090 and 3286 cm^{-1} confirmed the formation of metal free phthalocyanine complex (**2**). Generally infrared spectroscopy results indicate the presence of synthesized structures.

The ^1H NMR and ^{13}C NMR spectral datas of the synthesized compound (**1**) was found to be in accordance with the expected structure. In the ^{13}C NMR spectrum of (**1**), aromatic carbon atoms were observed approximately at the range of $165\text{--}118\text{ ppm}$. In the ^1H NMR spectra, the obtained signals for phthalocyanine (**2**) is quite broader compared to the signals of the starting 3-(4-(methylthio)phenoxy) phthalonitrile (**1**) derivative. The other resonances about aromatic protons and S- CH_3 protons in the ^1H NMR spectra of phthalocyanines (**2–4**) are very similar to that of the 3-(4-(methylthio)phenoxy) phthalonitrile. It should be mentioned that the other differences in the ^1H NMR spectra of phthalocyanine derivatives were the broad signals encountered in the case of compounds (**2,3** and **4**), owing to the aggregation of

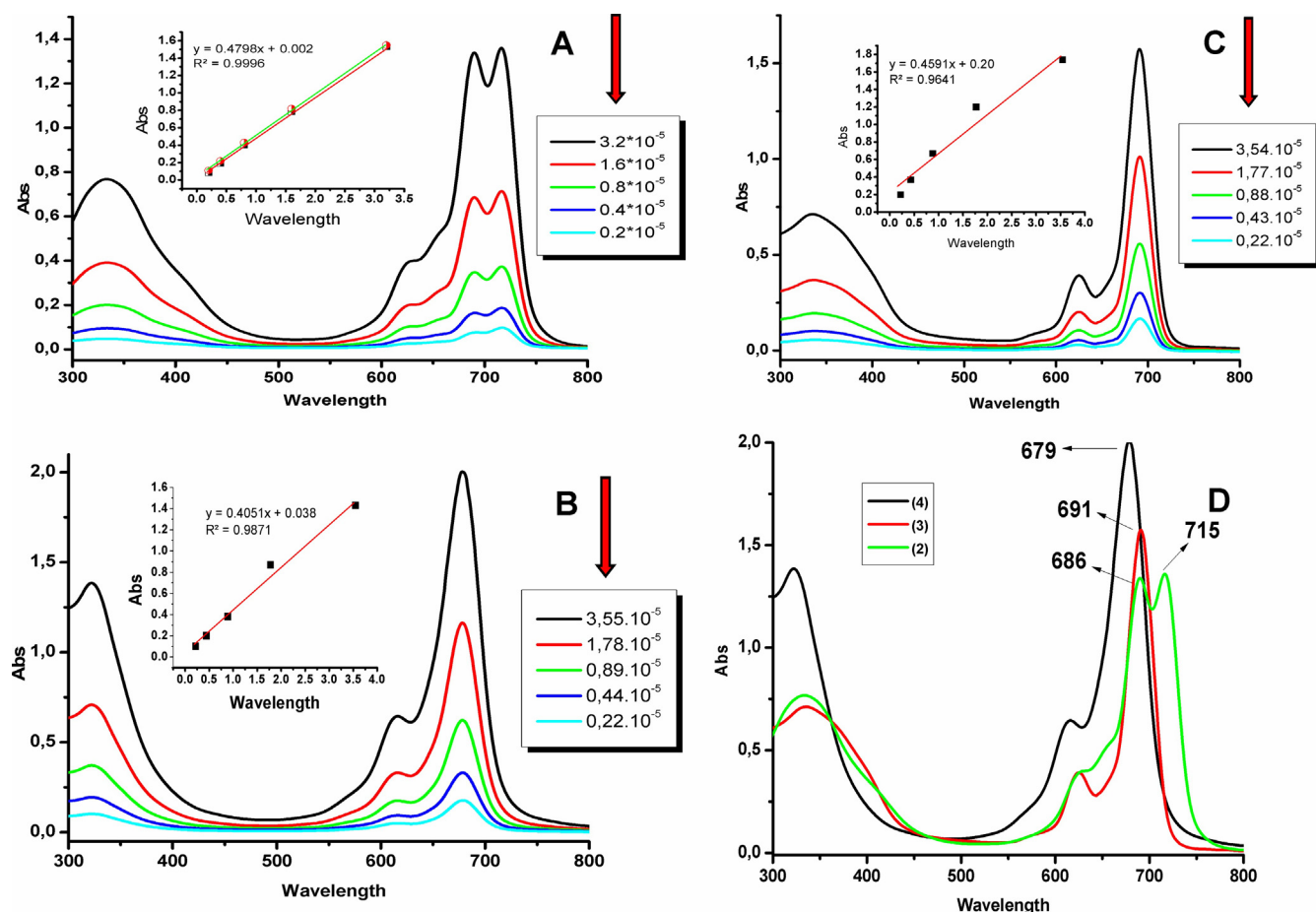


Fig. 1. Aggregation behaviors of A:(**2**), B:(**3**), C:(**4**) in THF at different concentrations and UV-vis spectra of the complexes D:(**2–4**).

planar phthalocyanine molecules at the considerably high concentration used for NMR measurements. They are in agreement with the structural information.

The MALDI-MS spectra of phthalocyanines (**2–4**) also confirmed the purposed structures of these phthalocyanines. The molecular ion peaks were observed at m/z : 1068.78 as $[M+H]^+$, 1132.58 as $[M+H]^+$ and 1125.38 as $[M+H]^+$, respectively.

UV-vis spectra of the phthalocyanine complexes (**2–4**) demonstrated characteristic absorptions in the Q-band region at around 650–700 nm, that is based on the π - π^* transition from the HOMO (highest occupied molecular orbital) to the LUMO (lowest unoccupied molecular orbital) of the Pc^{2-} ring, and in the B band region (UV region) at around 300–400 nm, resulting from the deeper π - π^* transitions. The Q-band absorptions of π - π^* transition for all phthalocyanines (**2–4**) in THF were observed as a single band of high intensity at 689(Q_x), 715(Q_y) for (**2**), 691 nm for (**3**), 679 nm for (**4**). There was also a shoulder at the slightly higher energy side of the Q band for all phthalocyanine complexes (**2–4**) (Fig. 1D).

3.2. Aggregation studies

The absorption spectra of the metal free phthalocyanine (**2**) and its metallo derivatives (**3**) and (**4**) in THF are typical for non-aggregated phthalocyanines which proves an intense and sharp Q-band. The dilution studies were made to test whether or not the aggregation occurs for the complexes (**2–4**) (shown in Fig. 1. (A–C) for (**2–4**) in THF at various concentrations). When the concentration increased, the intensity of the Q band also increased. There was no observed to be a new band because of the formation of aggregated species [19]. This indicated that the complexes (**2–4**) did not make any aggregation in THF and the Beer Lambert law was perfectly obeyed for all the complexes (**2–4**) for the concentrations between roughly 3.50×10^{-5} M and 0.20×10^{-6} M.

3.3. Metal ion binding titration studies

Attaching some functional groups containing sulfur atoms on the ring of the metal-free or metallo phthalocyanine complexes (MPcs) optically provides to discover or identify the presence of soft metal ions, such as Ag^+ and Pd^{2+} . Therefore, UV-vis spectroscopy was used to determine how the potential of the phthalocyanines is to detect the metal ions. In this study, to search for the possible aggregation behavior of the phthalocyanine complexes by metal ions, each titration experiment was performed by using a THF solution of (**2**) and (**3**), to make sure entire dissolution of the analyte salts, and $AgNO_3$ or Na_2PdCl_4 . The concentrations of all phthalocyanine complexes and the metal salt were prepared between roughly 2.5×10^{-5} and 1×10^{-3} M, respectively. The reason for the higher concentration of the metal salt was to decrease the spectral changes because of the volume changes during the titration.

Step-by-step, the treatment of (**2**) in THF with $AgNO_3$ in MeOH (in μ l portions) at room temperature firstly gave rise to green color to blue-green color and then a dark blue-precipitate started to occur at the bottom of the cuvette (Fig. 3A). These spectral changes may be attributed to the formation of H-aggregates (face-to-face), which was promoted by the treatment of $AgNO_3$ in MeOH. Absorbance at the Q-band of (**2**) in THF decreased with increasing $AgNO_3$ in MeOH. And a new band with a great blue-shift of 50 nm (around 636 nm) appeared (Fig. 2.A). At the situation of the treatment of complex (**3**) gradually with $AgNO_3$ in MeOH (in μ l portions) neither distinct shift to the longer wavelength nor to the short wavelength for both B and Q-bands appeared except for decreasing in the intensity in the UV-vis spectrum (Fig. 2C). The intensity of absorption of the Q band at 691 nm decreased. There

wasn't observed to be new band because of the formation of aggregated species. This proved that the complexes (**3**) did not show any aggregation/sensitivity in THF with increasing $AgNO_3$ content.

Upon the treatment of complex (**2**) in THF gradually with Na_2PdCl_4 in MeOH (in μ l portions) at room temperature lead to a decrease of Q_x and Q_y -bands belong to (**2**) at 686 and 715 nm and a new band in 636 nm occurred (Fig. 2B), suggesting H-aggregates (face-to-face) formation (Fig. 3C). On the other hand, the gradual treatment of (**3**) in THF with Na_2PdCl_4 in MeOH (in μ l portions) at room temperature gave rise to a similar color change, suggesting J-aggregates (head-to-tail) formation. The intensity of absorption at 691 nm gradually started to decrease with increasing Na_2PdCl_4 in MeOH (in μ l portions) and the new red-shifted band gradually appeared at 736 nm (Fig. 2D). These distinctive changes could be attributed to J-aggregates (head-to-tail) (Fig. 3E).

3.4. Fluorescence quantum yields

Fluorescence quantum yields were computed with the aid of Eq. (1) below,

$$\Phi_s = \Phi_{ref} \left(\frac{F_s}{F_{ref}} \right) \left(\frac{A_{ref}}{A_s} \right) \left(\frac{n_s^2}{n_{ref}^2} \right) \quad (1)$$

Where F_s and F_{ref} are area under the fluorescence emission spectra calculated of sample and standard, respectively, A_s and A_{ref} are absorbances at the same excitation wavelength of sample and standard, respectively, n_s and n_{ref} are refractive indexes of solvent used for sample and standard and Φ_{ref} is quantum yield of standard. ZnPc in DMSO ($\Phi_F = 0.20$) [20], ZnPc in THF ($\Phi_F = 0.32$) [21], for Q-band region were used as the standards. Both the sample and standard were excited at the same wavelength.

The fluorescence quantum yield (Φ_F) results of metal-free (**2**) and zinc (II) (**3**) phthalocyanine compounds were determined in the solvents of THF and DMSO. Fluorescence emission peaks were monitored for (**2**); at: 730 nm in THF and 734 nm in DMSO and for (**3**); at: 704 nm in THF and 712 nm in DMSO (Table 1). The obtained Stokes shifts for (**2**) and (**3**) are 15 and 13 in THF and 18 and 16 in DMSO, respectively (Fig. 4). The observed results are listed in Table 1. The Φ_F results of the studied phthalocyanine compounds (**2**) and (**3**) are lower than those of unsubstituted zinc (II) phthalocyanine (ZnPc) in THF and DMSO. The decrease in the Φ_F results is because of the substituents. This implies that the existence of non-peripheral 4-(methylthio)phenol substituents caused some fluorescence quenching of the parent (**2**) and (**3**). The fluorescence quantum yield Φ_F results for both (**2**) and (**3**), were low compared to unsubstituted ZnPc (Table 1). Especially, The lowest fluorescence quantum yields were obtained for both (**2**) and (**3**) in DMSO compared to the solvent of THF. The Φ_F values of the Zn (II) Pc compound (**3**) are higher than for the metal-free Pc compound (**2**) [22]. It is known that metal free phthalocyanines give low Φ_F values with the values actually improving on insertion of a diamagnetic central metal [23].

The excitation spectra and absorption spectra were very alike to each other and both were mirror images of the fluorescent spectra for (**3**) in THF and DMSO. The nearness of the wavelength of the Q band absorption maxima to the Q band maxima of the excitation spectra for (**3**) proved that the nuclear configurations of the ground and excited states are alike and not influenced by excitation [24]. But, the same situation wasn't observed for (**2**). The metal free phthalocyanine (**2**) did not show an entirely splitting and broadening of the Q band on excitation (Fig. 4B), providing the excitation spectrum unlike from the absorption spectrum. Since there was some aggregation for (**2**) even in the solvent of DMSO, there was a disharmony between absorption and excitation spectra

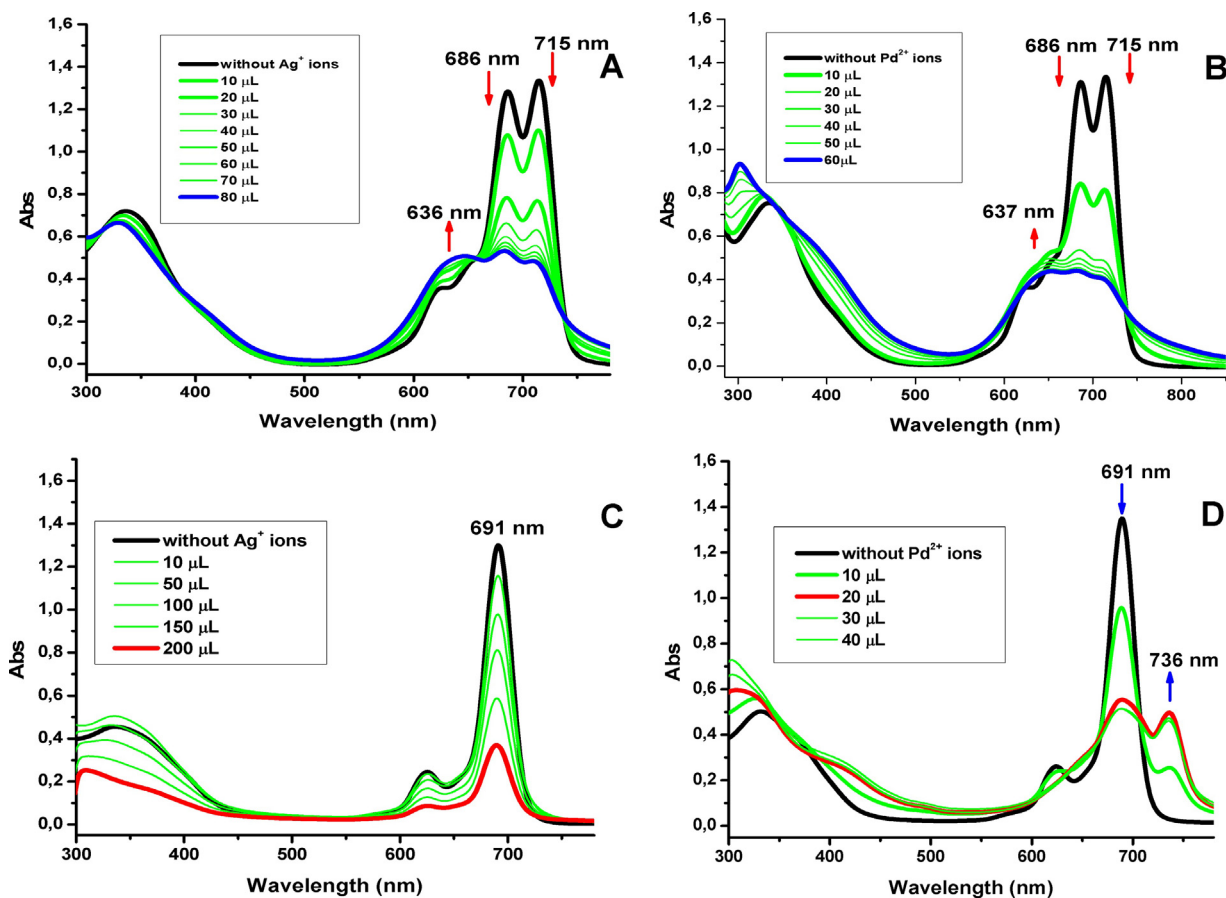


Fig. 2. UV-vis spectral change of **(2)** (A) and **(3)** (C) in THF during the titration by Ag^+ ions and **(2)** (B) and **(3)** (D) in THF during the titration by Pd^{2+} ions in MeOH.

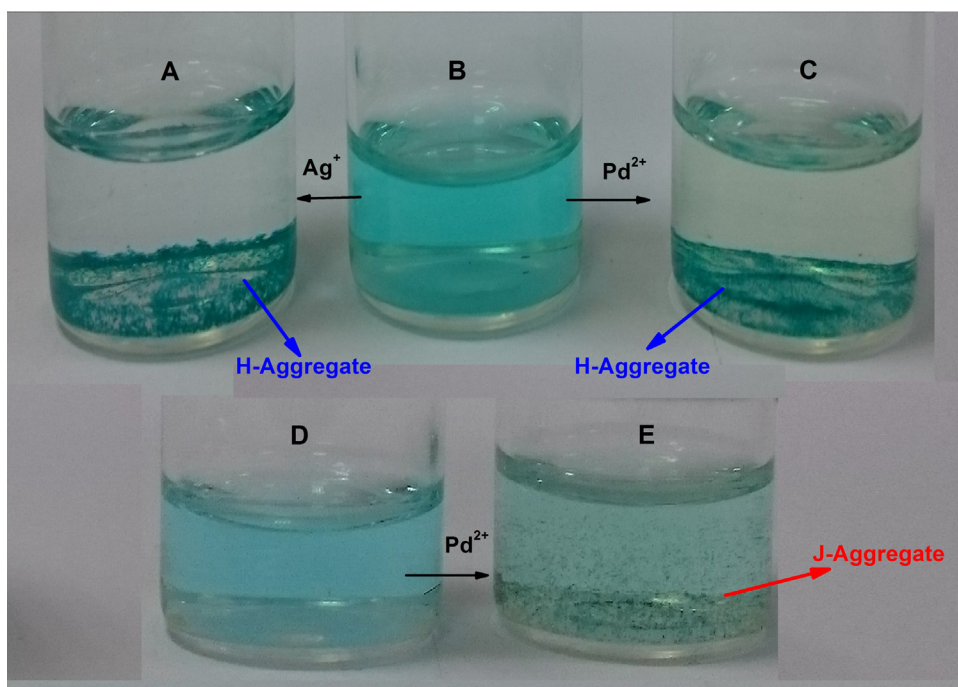


Fig. 3. A photograph of **(2)** in THF (2.5×10^{-5} M) in the absence (B) and presence of (A) Ag^+ ions (1×10^{-3} M) and (C) Pd^{2+} ions (1×10^{-3} M) and a photograph of **(3)** in THF (2.5×10^{-5} M) in the absence (D) and presence of (E) Pd^{2+} ions (1×10^{-3} M). The images were taken 12 h after the addition of Ag^+ and Pd^{2+} ions.

Table 1
Spectral parameters of the complexes (**2**) and (**3**) in THF and DMSO.

Compound	Q band, λ_{max} (nm)	$\log \varepsilon$	Excitation, λ_{Ex} (nm)	Emission, λ_{Em} (nm)	Stokes shift, ΔStokes (nm)	Fluorescence spectra and quantum, Φ_F
(2) in THF	686,715	4.61,4.63	715	730	15	0,055
(2) in DMSO	693,716	4.59,4.60	722	734	18	0,016
(3) in THF	691	4.65	690	704	13	0,118
(3) in DMSO	696	4.62	697	712	16	0,101
H ₂ Pc ^a in THF	691,722	4.99,4.93	700,729	749	27	0.045 ^a
ZnPc ^a in THF	699	5.07	699	719	20	0.121 ^a
Std ZnPc in DMSO	672 ^b	5.14 ^b	672 ^b	682 ^b	10 ^b	0.20

^a Data from Ref. [43].

^b Data from Ref. [44].

because aggregates do not fluoresce. The emission spectrum of the complex (**2**) is not split yet which is specific for metal free phthalocyanines that fluoresce with only one main peak in organic solvents [25,26]. Attaching functional groups at the non-peripheral region of phthalocyanine ring, with a greater destabilization effect of the HOMO level, is induced by greater linear combination of atomic orbitals (LCAO). This gives rise to a decreased HOMO-LUMO energy gap and a bathochromic shift occurs [27,28].

3.5. Electrochemical, optical and photovoltaic studies

The electrochemical behaviors of Pcs (**2**), (**3**), and (**4**) were investigated by CV and SWV in DCM/TBAPF₄. Figs. 5, 6 and 7. show

the CV and SWV curves and Table 2 lists corresponding electrochemical parameters of these Pcs including $E_{1/2}$, ΔE_p , and I_{pa}/I_{pc} values. Fig. 5.a shows the CV curve of (**2**), which indicates two oxidation and three reduction reactions. It is well known that all redox processes of (**2**) are assigned to the metal-free phthalocyanine ring [29]. While all the reduction reactions (R₁–R₃) are electrochemically reversible, the oxidation processes (O₁ and O₂) are quasi-reversible with respect to ΔE_p and I_{pa}/I_{pc} values. Moreover, the peak currents varied as a linear function of the square root of the scan rates ($\nu^{1/2}$) suggesting the diffusion-controlled redox behaviors of all the reduction reactions [30]. SWV curve also confirms the reversibility due to R₁–R₃ redox couples have symmetric with the same peak currents (Fig. 5b).

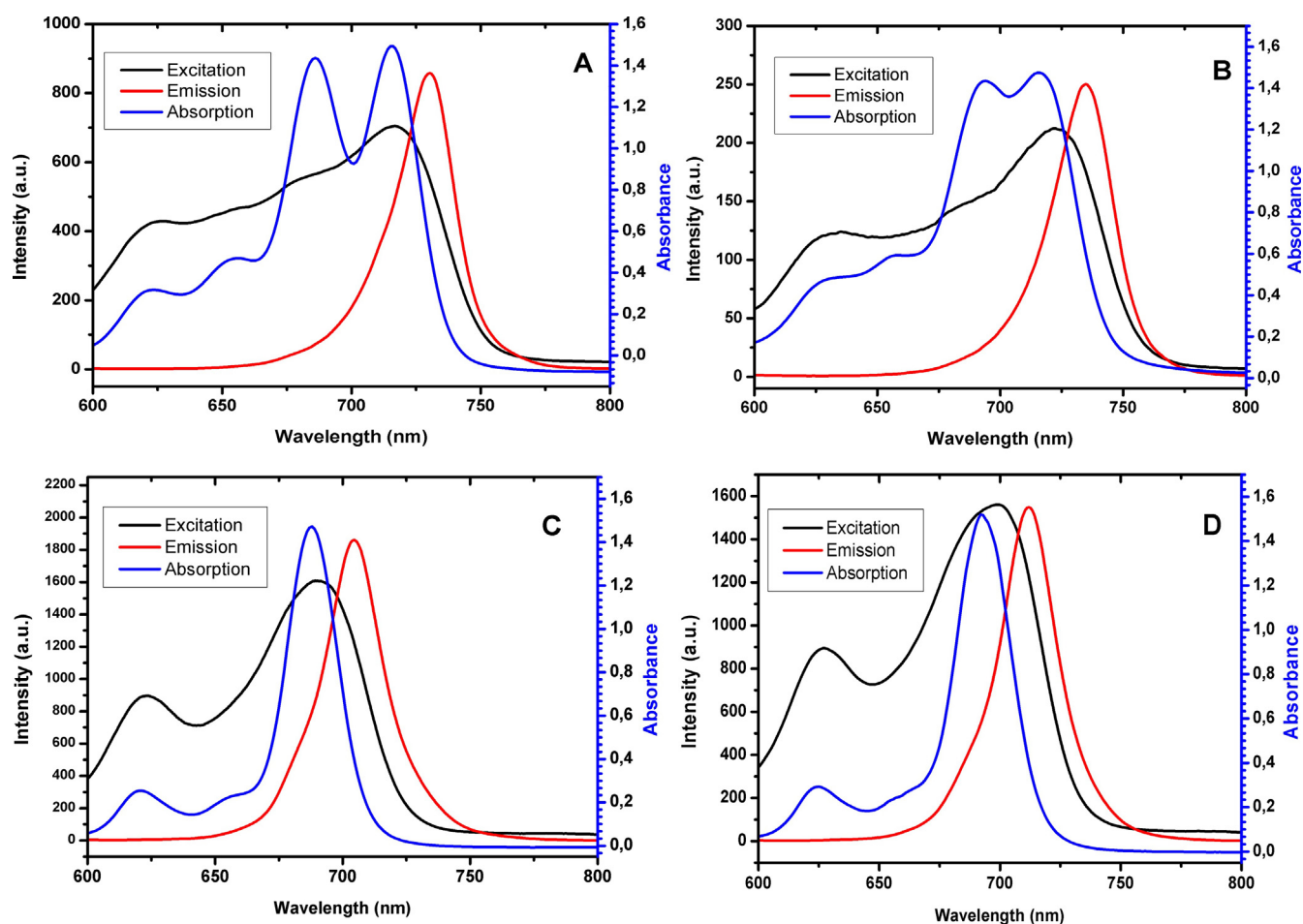


Fig. 4. Electronic absorption, excitation, and emission spectra of metal-free phthalocyanine (**2**) in THF (A) and DMSO (B) and zinc (II) phthalocyanine (**3**) in THF (C) and DMSO (D), respectively.

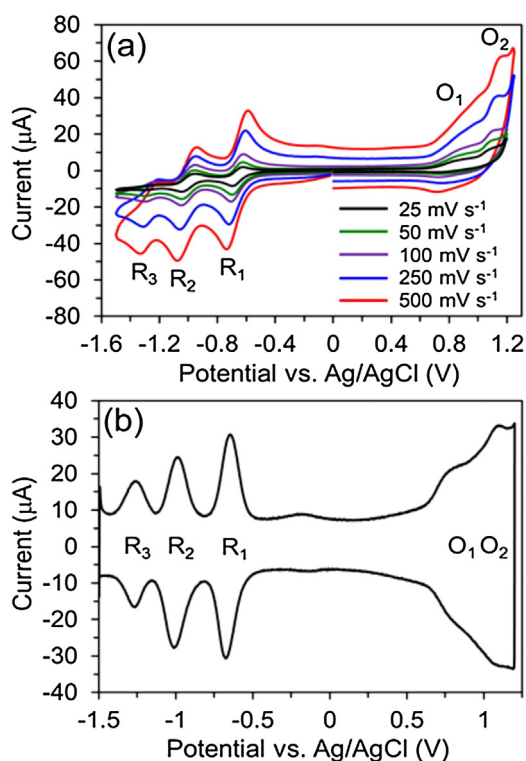


Fig. 5. (a) CVs of Pc (2) at various scan rates and (b) SWVs of Pc (2) recorded at 0.100 Vs^{-1} scan rate.

As shown in Fig. 6, all the redox couples for complex (3) are electrochemically reversible, except for the second reduction couple at -1.02 V ΔE_p and I_{pa}/I_{pc} values of this redox couple is

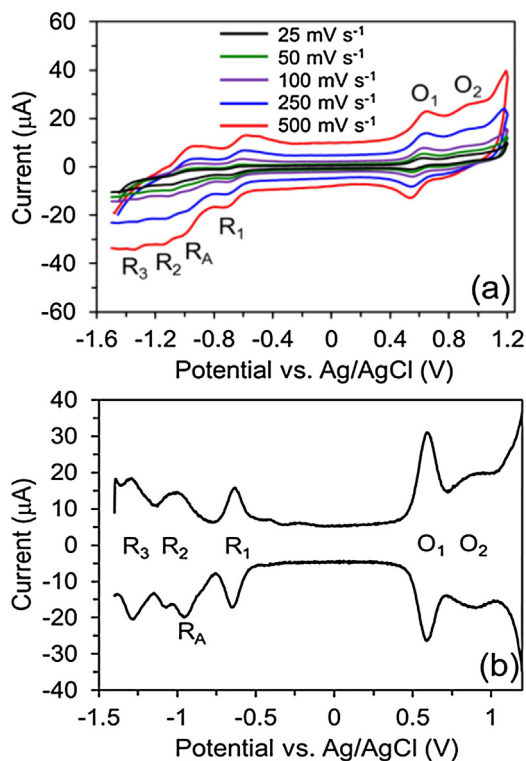


Fig. 6. (a) CVs of Pc (3) at various scan rates and (b) SWVs of Pc (3) recorded at 0.100 Vs^{-1} scan rate.

higher than that of reversible Fc/Fc^+ , whose ΔE_p and I_{pa}/I_{pc} is 109 mV and ~ 1.00 at 0.100 V s^{-1} scan rate, respectively. Obviously, cathodic process of the second couple splits into two peaks due to electron transfer to both the aggregated (R_A) and monomeric species (R_2). The R_A peak is also invisible at slow scan rates due to the dissociation of the aggregates at long times. As a consequence, the second redox couple exhibits poor reversibility due to the aggregation decreases the solubility, which makes the diffusion of molecules toward the electrode very slow for the cathodic process [31]. The slow diffusion of molecules decreases the cathodic R_2 peak current. The poor reversibility and aggregation was also clearly observed in the SWV curve (Fig. 6.b). However, the voltammetric behavior of (3) is very similar to that of (2), except the redox couples of (3) former slightly shifts to the negative potentials compared with that of (2). This similarity indicates that the redox processes for (3) involve the phthalocyanine ring rather than the redox-inactive metal center [7]. On the other hand, the potential shifting is due to the effective nuclear charge difference between the H^+ and Zn^{2+} ions [32].

The CV and SWV curves of complex (4) show two oxidation and three reduction couples, which is the same number to that of other Pcs, whereas the peak potentials of the reduction couples are different than those of the others (Fig. 7). Among them, especially the first reduction process (R_1) is significantly shifts toward positive potentials compared with those of (2) and (3). As known that the electrochemical properties of Co Pcs differ from those of metal-free and redox-inactive metal center phthalocyanines due to the energy levels of Co as redox-active metal center lie between the HOMO and the LUMO of the phthalocyanine ring. In other words, the metal reduction or oxidation occurs at potentials between those corresponding to reduction or oxidation of the ring in the Co Pcs [8,9]. Therefore, the R_1 process could be assigned to metal-based redox couple and the other processes to the ring. It can also be seen that the R_1 , R_2 and O_2 redox couples are reversible while the

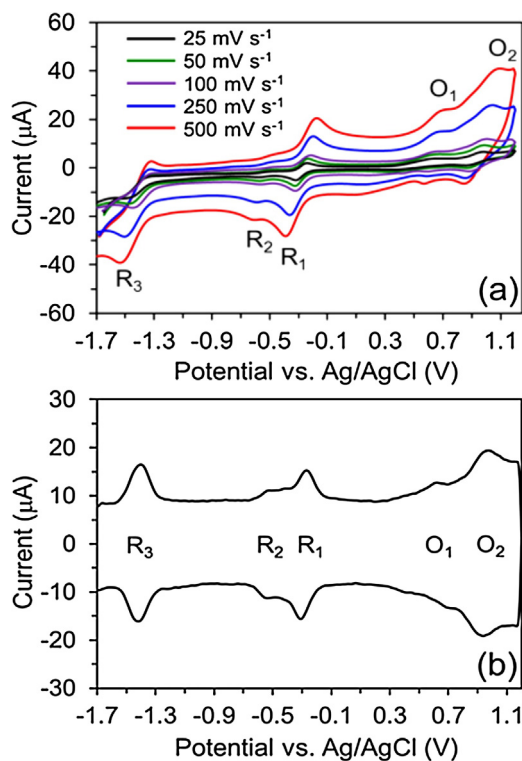


Fig. 7. (a) CVs of Pc (4) at various scan rates and (b) SWVs of Pc (4) recorded at 0.100 V s^{-1} scan rate.

Table 2
Voltammetric data of the Pcs. Potentials are reported vs. Ag/AgCl.

Dye	Parameters	Redox processes				
		O ₂	O ₁	R ₁	R ₂	R ₃
2	$E_{1/2}$ (V) ^a	1.09	0.78	−0.65	−0.99	−1.27
	ΔE_p (mV) ^b	119	131	89	91	81
	I_{pa}/I_{pc} ^c	1.14	0.74	0.98	0.95	0.96
3	$E_{1/2}$ (V) ^a	0.90	0.59	−0.66	−1.02	−1.29
	ΔE_p (mV) ^b	103	75	86	149	65
	I_{pa}/I_{pc} ^c	0.92	1.09	1.04	1.61	0.91
4	$E_{1/2}$ (V) ^a	0.96	0.62	−0.27	−0.56	−1.40
	ΔE_p (mV) ^b	104	51	87	75	123
	I_{pa}/I_{pc} ^c	0.93	1.41	0.96	0.89	0.68

^a $E_{1/2} = (E_{pa} + E_{pc})/2$ at 0.100 V s^{−1}.^b $\Delta E_p = E_{pa} - E_{pc}$ at 0.100 V s^{−1}.^c I_{pa}/I_{pc} for reduction, I_{pc}/I_{pa} for oxidation processes at 0.100 V s^{−1}.

R₃ process is quasi-reversible with respect to ΔE_p and I_{pa}/I_{pc} values. However, the ΔE_p value of O₁ process is lower than that of Fc/Fc⁺, which indicates reversible process, whereas its I_{pa}/I_{pc} value is higher than unity. This may be result from the presence of a chemical reaction succeeding the cathodic process of O₁ couple, since the chemical reaction decreases the cathodic current [33,34]. The SWV curve also supports the poor reversibility for O₁ process (Fig. 7a).

To investigate the electron injection and dye regeneration capability of the synthesized Pc dyes, related electrochemical and optical parameters were calculated, and corresponding data are listed in Table 3. Obviously, the HOMO levels of all the dyes are more positive than that of I[−]/I₃[−] redox potential value (0.4 V vs. NHE) [35], ensuring sufficient driving force for dye regeneration. On the other hand, all the LUMO levels are more negative than the TiO₂ conduction band energy level (−0.5 V vs. NHE) [35], which indicate that the excited state dyes can inject electrons into the conduction band of TiO₂.

In order to evaluate the photovoltaic performances, the DSSCs based on the Pcs as sensitizers were performed under standard simulated solar irradiation. However, the power conversion efficiencies ($\square$) for DSSCs based on Pcs (2) and (4) (data not shown here) were very lower than that of complex (3). The negligible efficiency for Pc (2) can be attributed to its lower LUMO energy level of the former Pc compared to that of (3) (Table 3). It is well known that the low LUMO levels cause insufficient electron flow from excited dyes to the conduction band of TiO₂ [36]. In the case of complex (4), the electron transfer may be different than that of (3) due to the energy level of Co in the core of ring lies between the HOMO and the LUMO of the Pc ring, which differs

Table 3
Electrochemical and optical properties of the Pc dyes.

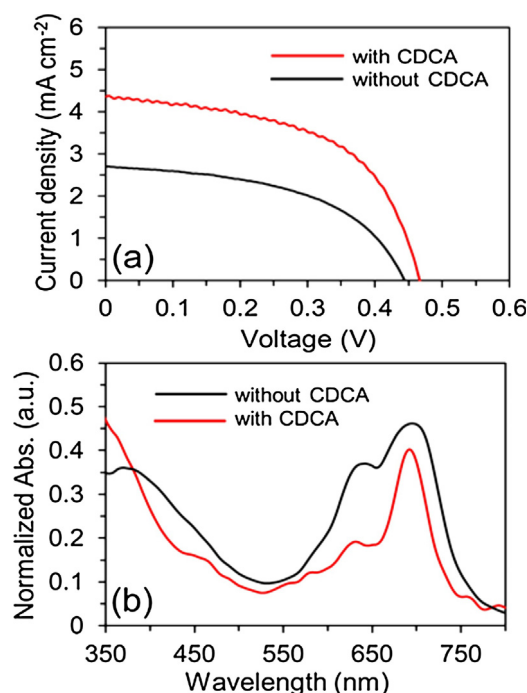
Dye	HOMO (V vs. NHE) ^a	E ₀₋₀ (eV) ^b	LUMO (V vs. NHE) ^c
2	0.87	1.68	−0.81
3	0.68	1.73	−1.05
4	0.71	1.73	−1.02

^a as an external reference, and its $E_{1/2}$ was found to be 0.54 V, vs. Ag/AgCl. By comparing this value with that of 0.63 V vs. NHE, the potentials vs. Ag/AgCl can be converted to that vs. NHE by adding a value of 0.09 V.

^b The HOMO was taken from of the first oxidation potential ($E_{1/2}$) in Table I. The potentials were calibrated with Fc/Fc⁺.

^c The band gap (E_{0-0}) was estimated from the absorption onset wavelength (λ_{onset}) using $E_{0-0} = 1240/\lambda_{onset}$.

^d The LUMO was calculated with the formula LUMO = HOMO − E_{0-0} .

**Fig. 8.** (a) The J-V curves of the DSSCs based on the complex (3) and (b) UV-vis absorption spectra of their corresponding films on TiO₂ surface.

from that of complex (3) [4]. Fig. 8a shows the J-V curves of the DSSCs based on complex (3) fabricated in the absence and presence of CDCA and the corresponding photovoltaic parameters are listed in Table 4. The DSSC fabricated with CDCA exhibits a high cell performance with a η of 1.09% ($J_{sc} = 4.37 \text{ mA cm}^{-2}$; $V_{oc} = 0.47 \text{ V}$; FF = 0.53) exceed that of without CDCA (0.61%). The increase in the J_{sc} can be attributed to the suppression of dye aggregation in the assistance of CDCA [37], while the increase in V_{oc} can be ascribed to the reduction of charge recombination by the formation of a thin layer of CDCA on the free TiO₂ surface [38].

Aggregation of complex (3) on TiO₂ surface was also confirmed by UV-vis absorption measurements (Fig. 8b). The absorption spectrum of (3) without CDCA exhibits a broadening Q-band absorption as well as a new band at around 633 nm, clearly indicating the aggregation of the dyes on the TiO₂ surface [39]. However, the width of the Q-band and the absorption intensity of the new band remarkably decreased in the presence of CDCA, which indicates that the aggregates on the TiO₂ surface were dissociated by co-adsorption with CDCA [40,41].

On the other hand, the complex (3) as sensitizer has any anchoring groups such as carboxylic acid or pyridine for the formation of coordinate bonds between the Pc molecules and the semiconductor surface. However, when considering the fact that the DSSC based on the ZnPc can convert solar energy into electricity, it can be expected to the presence of a physical adsorption between electron-donating oxygen atoms via non-bonding electrons in the molecule and positive charged Ti⁴⁺ ions in the TiO₂ [12,42].

Table 4
Photovoltaic parameters of the DSSCs.

Dye	CDCA (mM)	J_{sc} (mA cm ^{−2})	V_{oc} (V)	FF	η (%)
3	0.00	2.72	0.44	0.51	0.61
3	1.00	4.37	0.47	0.53	1.09

4. Conclusion

We have synthesized non peripherally tetra 4-(methylthio) phenoxy substituted metal-free (2), Zn(II) (3) and Co(II) (4) phthalocyanine and characterized by elemental analysis, FT-IR, ¹H NMR, ¹³C NMR, MALDI-MS, UV–vis spectral data. Before and after the treatment of phthalocyanines with soft metal Ag⁺ or Pd²⁺ ions, the complexes (2) and (3) were investigated by UV–vis spectral changes if there happens aggregation or not, and at the same time what kind of aggregation if there happens. Fluorescence quantum yields (Φ_F) of phthalocyanine (3) were found to be higher than metal free phthalocyanine (2) that is typical for phthalocyanine complexes. Also, the lower Φ_F values were obtained for the phthalocyanines (2) and (3) than unsubstituted ZnPc in the solvents used (THF and DMSO). Voltammetric studies revealed that all the complexes exhibit electrochemical reversible or quasi-reversible redox behaviors. The electrochemical, optical and photovoltaic properties of all the phthalocyanines were examined for their using in DSSCs. The DSSC fabricated with complex (3) showed a conversion efficiency of 1.09%, whereas the Pcs (2) and (4) did not work in their corresponding DSSCs due to lower LUMO energy level of (2) and the different energy level of Co in (4) than those that of (3).

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References

- [1] C.C. Leznoff, A.B.P. Lever (Eds.), *Phthalocyanines: Properties and Applications*, 1–4, VCH, Weinheim, 1989 1989.
- [2] A. Kalkan, A. Aytekin, Z.A. Bayır, *Metallophthalocyanines with rod-shaped substituents*, *Trans. Met. Chem.* 31 (2006) 720–723.
- [3] Z. Biyikoglu, S.Z. Yıldız, Kantekin H. Metal-free, metallophthalocyanines appending with eight 12-crown-4 ethers, *J. Org. Chem.* 695 (2010) 1729–1733.
- [4] C.C. Leznoff, S.M. Marcuccio, S. Greenberg, A.B.P. Lever, K.B. Tomer, *Metallophthalocyanine dimers incorporating 5-atom covalent bridges*, *Can. J. Chem.* 63 (3) (1985) 623–631.
- [5] A.T. Bilgiçli, M. Durdasoglu, E. Kirbaş, M.N. Yarasir, M. Kandaz, *Metal ion sensing soluble a or b tetrasubstituted gallium and indium phthalocyanines: synthesis, characterization, photochemistry and aggregation behaviors*, *Polyhedron* 100 (2015) 1–9.
- [6] N. Akkaya, A.T. Bilgiçli, A. Aytekin, M.N. Yarasir, M. Kandaz, *Novel metal(III) and metal free soft phthalocyanine metal ion sensors bearing (1-hydroxyhexan-3-ylthio)-substituents: synthesis, characterization, aggregation behavior*, *Polyhedron* 85 (2015) 857–863.
- [7] A. Gouloumis, S.G. Liu, A. Sastre, P. Vazquez, L. Echegoyen, T. Torres, *Synthesis and electrochemical properties of phthalocyanine-Fullerene hybrids*, *Chem. Eur. J.* 6 (19) (2000) 3600–3607.
- [8] J.H. Zagal, M.A. Gulppi, G. Cardenas-Jiron, *Metal-centered redox chemistry of substituted cobalt phthalocyanines adsorbed on graphite and correlations with MO calculations and Hammett parameters. Electrocatalytic reduction of a disulfide*, *Polyhedron* 19 (22–23) (2000) 2255–2260.
- [9] E. Guzel, A. Koca, M.B. Kocak, *Anionic water-soluble sulfonated phthalocyanines: microwave-assisted synthesis, aggregation behaviours, electrochemical and in-situ spectroelectrochemical characterization*, *Supramol. Chem.* 29 (7) (2017) 536–546.
- [10] G. Gumrukcu, M.U. Ozgur, A. Altindal, A.R. Ozkaya, B. Salih, O. Bekaroglu, *Synthesis and electrochemical, electrical and gas sensing properties of novel mononuclear metal-free, Zn(II), Ni(II), Co(II), Cu(II), Lu(III) and double-decker Lu(III) phthalocyanines substituted with 2-(2H-1,2,3-benzotriazol-2-yl)-4-(1,1,3,3-tetramethylbutyl) phenoxy*, *Synth. Met.* 161 (1–2) (2011) 112–123.
- [11] H. Imahori, T. Umeyama, K. Kurotobi, Y. Takano, *Self-assembling porphyrins and phthalocyanines for photoinduced charge separation and charge transport*, *Chem. Commun.* 48 (34) (2012) 4032–4045.
- [12] H.E.A. Ali, A. Altindal, S. Altun, Z. Odabas, *Highly efficient dye-sensitized solar cells based on metal-free and copper(II) phthalocyanine bearing 2-phenylphenoxy moiety*, *Dyes Pigm.* 124 (2016) 180–187.
- [13] T. Ikeuchi, H. Nomoto, N. Masaki, M.J. Griffith, S. Mori, M. Kimura, *Molecular engineering of zinc phthalocyanine sensitizers for efficient dye-sensitized solar cells*, *Chem. Commun.* 50 (16) (2014) 1941–1943.
- [14] T. Komori, Y. Amao, *Dye-sensitized solar cell with the near-infrared sensitization of aluminum phthalocyanine*, *J. Porphyrins Phthalocyanines* 7 (2) (2003) 131–136.
- [15] N. Li, Z.X. Sun, R. Liu, L. Xu, K. Xu, X.M. Song, *Enhanced power conversion efficiency in phthalocyanine-sensitized solar cells by modifying TiO₂ photoanode with polyoxometalate*, *Sol. Energy Mater. Sol. Cells* 157 (2016) 853–860.
- [16] M. Sevim, M.N. Yarasir, A. Koca, M. Kandaz, *Novel scorpion type phthalocyanine chemosensors for detection of selective-metal ion by inducing H- and J-aggregations in solution; synthesis, characterization and electrochemistry*, *Dyes Pigm.* 111 (2014) 190–201.
- [17] S. Tuncer, A. Koca, A. Gul, U. Avciata, *1,4-Dithiaheterocycle-fused porphyrazines: synthesis, characterization, voltammetric and spectroelectrochemical properties*, *Dyes Pigm.* 81 (2) (2009) 144–151.
- [18] B.X. Lei, Q.P. Luo, X.Y. Yu, W.Q. Wu, C.Y. Su, D.B. Kuang, *Hierarchical TiO₂ flowers built from TiO₂ nanotubes for efficient Pt-free based flexible dye-sensitized solar cells*, *Phys. Chem. Chem. Phys.* 14 (38) (2012) 13175–13179.
- [19] Y. Zorlu, F. Dumoulin, M. Durmus, V. Ahsen, *Comparative studies of photophysical and photochemical properties of solketal substituted platinum(II) and zinc(II) phthalocyanine sets*, *Tetrahedron* 66 (2010) 3248–3258.
- [20] A. Ogunsipe, J.Y. Chen, T. Nyokong, *Photophysical and photochemical studies of zinc(II) phthalocyanine derivatives – effects of substituents and solvents*, *New J. Chem.* 7 (2004) 822–827.
- [21] P. Zimcik, V. Novakova, K. Kopecky, M. Miletin, R.Z. Uslu Kobak, E. Svandrikova, et al., *Magnesium azaphthalocyanines: an emerging family of excellent redemitting fluorophores*, *Inorg. Chem.* 51 (2012) 4215–4223.
- [22] M. Çamur, M. Durmus, M. Bulut, *Synthesis, characterization and comparative studies on the photophysical and photochemical properties of metal-free and zinc(II) phthalocyanines with phenyloxycetic acid functionalities*, *Polyhedron* 12 (30) (2011) 1935–1944.
- [23] T. Nyokong, E. Antunes, in: K.M. Kadish, K.M. Smith, R. Guilard (Eds.), *The Handbook of Porphyrin Science*, 7, Academic Press/World Scientific, Singapore, 2010, pp. 247–358.
- [24] M. Özçesmeçi, *Synthesis, photophysical and photochemical properties of metal-free and zinc(II) phthalocyanines bearing alpha-naphtholbenzein units*, *J. Org. Chem.* 767 (2014) 16–21.
- [25] M. Özçesmeçi, A. Gelir, A. Gül, *Synthesis and photophysical properties phthalocyanine-pyrene dyads*, *Dyes Pigm.* 92 (2012) 954–960.
- [26] M. Özçesmeçi, O. Güney, A.I. Okur, A. Gül, *New phthalocyanines containing bulky electron rich substituents*, *J. Porphyr. Phthalocyanines* 13 (2009) 753–759.
- [27] N.B. McKeown, H. Li, M. Helliwell, *A non-planar, hexadeca-substituted, metal-free phthalocyanine*, *J. Porphyrins Phthalocyanines* 9 (2005) 841–845.
- [28] A.B. Anderson, T.L. Gorden, M.E. Kenney, *Electronic and redox properties of stacked-ring silicon phthalocyanines from molecular-orbital theory*, *J. Am. Chem. Soc.* 107 (1985) 192–195.
- [29] I. Omeroglu, T. Arslan, Z. Biyikoglu, G. Tosun, *Novel phthalocyanines bearing 4-ferrocenylphenoxy substituents and their electrochemistry*, *J. Organomet. Chem.* 749 (2014) 261–265.
- [30] P. Tau, T. Nyokong, *Electrochemical characterisation of tetra- and octa-substituted oxo(phthalocyaninato)titanium(IV) complexes*, *Electrochim. Acta* 52 (11) (2007) 3641–3650.
- [31] T.T. Tasso, T. Furuyama, N. Kobayashi, *Absorption and electrochemical properties of cobalt and iron phthalocyanines and their quaternized derivatives: aggregation equilibrium and oxygen reduction electrocatalysis*, *Inorg. Chem.* 52 (16) (2013) 9206–9215.
- [32] H.R.P. Karaoglu, A. Koca, M.B. Kocak, *Synthesis electrochemical and spectroelectrochemical characterization of novel soluble phthalocyanines bearing chloro and quaternizable bulky substituents on peripheral positions*, *Dyes Pigm.* 92 (3) (2012) 1005–1017.
- [33] K.C.D. Leite, L.M.F.C. Torres, L.F. Garcia, S.G. Rezende, J.R. de Oliveira-Neto, F.M. Lopes, T.A. Garcia, R.M. Verly, W.T.P. dos Santos, E.D. Gil, *Electrochemical characterization of scopoletin, a 7-hydroxy-6-methoxy-coumarin*, *Int. J. Electrochem. Sci.* 10 (7) (2015) 5714–5725.
- [34] E. Guzel, A. Koca, A. Gul, M.B. Kocak, *Microwave-assisted synthesis, electrochemistry and spectroelectrochemistry of amphiphilic phthalocyanines*, *Synth. Met.* 199 (2015) 372–380.
- [35] W.J. Ying, J.B. Yang, M. Wielopolski, T. Moehl, J.E. Moser, P. Comte, J.L. Hua, S.M. Zakeeruddin, H. Tian, M. Gratzel, *New pyrido[3,4-b] pyrazine-based sensitizers for efficient and stable dye-sensitized solar cells*, *Chem. Sci.* 5 (1) (2014) 206–214.
- [36] K.D. Seo, I.T. Choi, H.K. Kim, *Organic dyes with well-Defined structures for highly efficient dye-Sensitized solar cells based on a cobalt electrolyte*, *Chem. Eur. J.* 21 (42) (2015) 14804–14811.
- [37] J.H. Yum, S.J. Moon, R. Humphry-Baker, P. Walter, T. Geiger, F. Nuesch, M. Graetzel, M.D.K. Nazeeruddin, *Effect of coadsorbent on the photovoltaic performance of squaraine sensitized nanocrystalline solar cells*, *Nanotechnology* 19 (42) (2008) 424005.
- [38] A.C. Khazraji, S. Hotchandani, S. Das, P.V. Kamat, *Controlling dye (Merocyanine-540) aggregation on nanostructured TiO₂ films: an organized assembly approach for enhancing the efficiency of photosensitization*, *J. Phys. Chem. B* 103 (22) (1999) 4693–4700.
- [39] J.H. Yum, S.R. Jang, R. Humphry-Baker, M. Gratzel, J.J. Cid, T. Torres, M.K. Nazeeruddin, *Effect of coadsorbent on the photovoltaic performance of zinc phthalocyanine-sensitized solar cells*, *Langmuir* 24 (10) (2008) 5636–5640.
- [40] H. Matsuzaki, T.N. Murakami, N. Masaki, A. Furube, M. Kimura, S. Mori, *Dye aggregation effect on interfacial electron-Transfer dynamics in zinc*

- phthalocyanine-Sensitized solar cells, *J. Phys. Chem. C* 118 (31) (2014) 17205–17212.
- [41] S. Yamamoto, A. Zhang, M.J. Stillman, N. Kobayashi, M. Kimura, Low-Symmetry omega-Shaped zinc phthalocyanine sensitizers with panchromatic light-Harvesting properties for dye-Sensitized solar cells, *Chem. Eur. J.* 22 (52) (2016) 18760–18768.
- [42] N. Ishida, D. Fujita, Adsorption of Co-phthalocyanine on the rutile TiO₂ (110) surface: a scanning tunneling microscopy/spectroscopy study, *J. Phys. Chem. C* 116 (38) (2012) 20300–20305.
- [43] S. Tuncer, K. Kaya, I. Özçeşmeci, A. Kalkan, Non-peripherally tetrasubstituted phthalocyanines: Synthesis, characterization and, photophysical investigation, *J. Org. Chem.* 827 (2017) 78–85.
- [44] I. Gürol, M. Durmuş, V. Ahsen, T. Nyokong, Synthesis, photophysical and photochemical properties of substituted zinc phthalocyanines, *Dalton Trans.* 34 (2007) 3782–3791.